

Phishing Websites Dataset

Plain-English Guide to All 31 Columns (UCI Phishing Websites Dataset, 11,055 rows)

This dataset describes websites using 30 features (clues) that help tell phishing sites apart from legitimate ones, plus 1 label column (the answer). Every feature is encoded as a number: most use **-1 = suspicious/phishing sign** and **1 = safe/legitimate sign** (some also use **0 = suspicious, in-between case**). The final column, **Result**, is the label: -1 means the site is phishing, 1 means it is legitimate.

1. URL-Based Features (checked directly from the web address)

Feature	What it checks	Why it signals phishing
having_IP_Address	Checks if the URL uses a raw IP address instead of a domain name (e.g. http://192.168.1.1/login).	Phishers use IPs to avoid registering/paying for a traceable domain.
URL_Length	Measures how long the full URL is.	Phishers often use long URLs to hide suspicious parts or fake domains.
Shortening_Service	Checks if a URL-shortening service was used (like bit.ly, tinyurl).	Shorteners hide the real destination link from the user.
having_At_Symbol	Checks if the URL contains an '@' symbol.	Browsers ignore everything before '@', so attackers hide the real domain after it.
double_slash_redirecting	Checks if '/' appears later in the URL path (not just after http:).	Indicates the user may be silently redirected to another site.
Prefix_Suffix	Checks if the domain name contains a hyphen '-' (e.g. paypal-secure.com).	Legitimate domains rarely use hyphens; phishers add them to mimic brand names.
having_Sub_Domain	Counts how many sub-domains/dots are in the URL.	Too many sub-domains is a common trick to disguise the real site (e.g. paypal.com.fake.net).
SSLfinal_State	Checks if the site has a valid HTTPS/SSL certificate and its trust level.	Phishing sites often lack proper SSL certificates or use freshly issued/low-trust ones.
Domain_registration_length	Checks how long until the domain registration expires.	Phishing domains are usually registered for only 1 year (cheap, short-term use).
Favicon	Checks if the small browser tab icon (favicon) is loaded from the same domain.	Phishing pages often load the favicon from an external/legitimate site to appear trustworthy.
port	Checks if the URL uses non-standard network ports.	Unusual ports can indicate the connection is not going through normal secure channels.
HTTPS_token	Checks if the word 'https' is inserted into the domain name itself (e.g. https-paypal.com).	Attackers add 'https' as text to trick users into thinking the site is secure.

2. Abnormal-Based Features (checked from page content vs. domain)

Feature	What it checks	Why it signals phishing
Request_URL	Checks what percentage of images/videos/objects on the page are loaded from a different domain.	Legitimate sites load most resources from their own domain; phishing pages often pull content from elsewhere.
URL_of_Anchor	Checks what percentage of clickable links (anchors) point to a different domain.	Phishing pages often link out to unrelated or malicious domains.
Links_in_tags	Checks what percentage of meta, script, and link tags point to a different domain.	High percentage suggests the page content isn't actually hosted where it appears to be.
SFH	Checks the Server Form Handler - where does a login/data form actually submit to?	If a form submits to a blank page or a different domain, it's a major phishing red flag.
Submitting_to_email	Checks if the page submits form data directly to an email address (mailto:).	Legitimate sites use secure servers, not raw email, to collect form data.
Abnormal_URL	Checks if the URL matches the identity/host info in the site's WHOIS record.	A mismatch suggests the domain is being impersonated or misused.

3. HTML & JavaScript-Based Features (checked from page behavior)

Feature	What it checks	Why it signals phishing
Redirect	Counts how many times the page redirects before finally loading.	Phishing pages often chain multiple redirects to confuse tracking and users.
on_mouseover	Checks if the site changes the status bar text using onMouseOver (JavaScript).	Used to fake a legitimate-looking link/URL when a user hovers over it.
RightClick	Checks if right-click is disabled on the page.	Phishers disable right-click to stop users from viewing page source and detecting fakes.
popUpWidnow	Checks if the site uses pop-up windows that request user input.	Legitimate sites rarely ask for sensitive info via pop-ups; phishers do.
Iframe	Checks if the page uses invisible iframes.	Iframes can be used to overlay fake content or hide malicious elements.

4. Domain-Based Features (checked from domain history/reputation)

Feature	What it checks	Why it signals phishing
age_of_domain	Checks how old the domain is (time since registration).	Phishing domains are usually very new (weeks/months old), while legit sites are older.
DNSRecord	Checks whether the domain has a valid DNS record.	Phishing domains sometimes have no or broken DNS records since they're short-lived.
web_traffic	Checks the site's traffic/popularity ranking (e.g. via Alexa rank).	Phishing sites typically have little to no real traffic history.

Feature	What it checks	Why it signals phishing
Page_Rank	Checks the site's Google PageRank score.	Legitimate, established sites tend to have higher PageRank; phishing sites usually score very low.
Google_Index	Checks if the page is indexed by Google.	Most phishing pages are short-lived and never get indexed by search engines.
Links_pointing_to_page	Counts how many other websites link to this page (backlinks).	Legitimate sites accumulate backlinks over time; phishing pages usually have zero or very few.
Statistical_report	Checks if the domain/IP appears on known phishing/blacklist statistical reports (e.g. PhishTank, StopBadware).	A direct match with known-bad host lists is a strong phishing indicator.

5. Target Column (the label your model predicts)

Feature	What it checks	Why it signals phishing
Result	The final answer column: -1 = phishing website, 1 = legitimate website.	This is what your machine learning model is trying to predict using the 30 features above.

Quick Notes for Your Project

- All 30 features are already numerically encoded in the CSV (mostly -1/1, some -1/0/1) so no extra text-cleaning is needed.
- Categories to remember when explaining your project: URL-based, Abnormal-based, HTML/JavaScript-based, and Domain-based features.
- Strong individual predictors often cited in research on this dataset: SSLfinal_State, URL_of_Anchor, and web_traffic.
- Result is your target/label column: -1 = phishing, 1 = legitimate.
- Source: Mohammad, R., & McCluskey, L. (2012). Phishing Websites Dataset, UCI Machine Learning Repository.